

Homework Template

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Problem 1

Pick a box uniformly at random. Box 1 has 2 white, 3 blue. Box 2 has 4 white, 1 blue.

$$\Pr(B_1 | \text{blue}) = \frac{\Pr(\text{blue} | B_1) \Pr(B_1)}{\Pr(\text{blue})} = \frac{\frac{3}{5} \cdot \frac{1}{2}}{\frac{1}{2} \cdot \frac{3}{5} + \frac{1}{2} \cdot \frac{1}{5}} = \frac{3/10}{4/10} = \frac{3}{4}.$$

Problem 2

Prevalence $\Pr(I) = 0.04$. Sensitivity $\Pr(+ | I) = 0.97$. False positive rate $\Pr(+ | I^c) = 0.02$. Thus $\Pr(- | I) = 0.03$, $\Pr(- | I^c) = 0.98$, and $\Pr(I^c) = 0.96$.

1.

$$\Pr(I | +) = \frac{\Pr(+ | I) \Pr(I)}{\Pr(+ | I) \Pr(I) + \Pr(+ | I^c) \Pr(I^c)} = \frac{0.04 \cdot 0.97}{0.04 \cdot 0.97 + 0.96 \cdot 0.02} = \frac{97}{145} \approx 0.669.$$

2.

$$\Pr(I^c | +) = 1 - \Pr(I | +) = \frac{48}{145} \approx 0.331.$$

3.

$$\Pr(I | -) = \frac{\Pr(- | I) \Pr(I)}{\Pr(- | I) \Pr(I) + \Pr(- | I^c) \Pr(I^c)} = \frac{0.04 \cdot 0.03}{0.04 \cdot 0.03 + 0.96 \cdot 0.98} = \frac{1}{785} \approx 0.001274.$$

4.

$$\Pr(I^c | -) = 1 - \Pr(I | -) = \frac{784}{785} \approx 0.998726.$$

Problem 3

$\Pr(S = 1) = \frac{1}{3}$, $\Pr(S = 0) = \frac{2}{3}$. $\Pr(R = 0 | S = 0) = 0.9$, $\Pr(R = 1 | S = 0) = 0.1$. $\Pr(R = 1 | S = 1) = 0.8$, $\Pr(R = 0 | S = 1) = 0.2$.

1.

$$\Pr(R = 0) = \Pr(S = 0) \Pr(R = 0 | S = 0) + \Pr(S = 1) \Pr(R = 0 | S = 1) = \frac{2}{3} \cdot 0.9 + \frac{1}{3} \cdot 0.2 = \frac{2}{3}.$$

2.

$$\Pr(S = 0 | R = 0) = \frac{\Pr(S = 0) \Pr(R = 0 | S = 0)}{\Pr(R = 0)} = \frac{\frac{2}{3} \cdot 0.9}{\frac{2}{3}} = 0.9 = \frac{9}{10}.$$

3.

$$\Pr(S = 0 | R = 1) = \frac{\Pr(S = 0) \Pr(R = 1 | S = 0)}{\Pr(R = 1)} = \frac{\frac{2}{3} \cdot 0.1}{\frac{1}{3}} = 0.2 = \frac{1}{5}.$$

Problem 4

There are three cards: BB, RR, BR. Picking a card at random and observing one side is equivalent to choosing uniformly from the six sides.

1. Given that the observed side is black, the black sides available are: two from BB and one from BR, three in total. The opposite side is black only if the side came from BB.

$$\Pr(\text{other side black} \mid \text{seen black}) = \frac{2}{3}.$$

2. The probability that the opposite side matches the observed color equals the fraction of sides whose opposite matches: BB contributes 2 matching sides, RR contributes 2 matching sides, BR contributes 0. Out of 6 sides total,

$$\Pr(\text{opposite matches observed}) = \frac{4}{6} = \frac{2}{3}.$$

Problem 5

Let H be the number of heads in n flips of a fair coin.

$$\Pr(\text{equal heads and tails}) = \begin{cases} \frac{\binom{n}{n/2}}{2^n}, & \text{if } n \text{ is even,} \\ 0, & \text{if } n \text{ is odd.} \end{cases}$$

Problem 6

Assume $\Pr(A) \neq 0$ and $\Pr(B) \neq 0$. If $\Pr(B \mid A) < \Pr(B)$, then $\Pr(A \mid B) < \Pr(A)$.

Proof. By Bayes' rule,

$$\Pr(B \mid A) = \frac{\Pr(A \mid B) \Pr(B)}{\Pr(A)}.$$

If $\Pr(B \mid A) < \Pr(B)$, then

$$\frac{\Pr(A \mid B) \Pr(B)}{\Pr(A)} < \Pr(B).$$

Since $\Pr(B) > 0$, divide both sides by $\Pr(B)$ to get

$$\frac{\Pr(A \mid B)}{\Pr(A)} < 1 \implies \Pr(A \mid B) < \Pr(A).$$

□

Problem 7

Given $\Pr(A) = \frac{3}{4}$ and $\Pr(B) = \frac{1}{3}$.

1. Bounds for $\Pr(A \cap B)$:

$$\max \Pr(A \cap B) = \min\{\Pr(A), \Pr(B)\} = \frac{1}{3}, \quad \min \Pr(A \cap B) = \max\{0, \Pr(A) + \Pr(B) - 1\} = \frac{1}{12}.$$

Examples:

- Max $\frac{1}{3}$: take $B \subseteq A$ with $\Pr(B) = \frac{1}{3}$.

- Min $\frac{1}{12}$: on $[0, 1]$ let $A = [0, \frac{3}{4}]$, $B = [\frac{2}{3}, 1]$. Then $|A \cap B| = \frac{3}{4} - \frac{2}{3} = \frac{1}{12}$.

2. Using $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$,

$$\max \Pr(A \cup B) = \Pr(A) + \Pr(B) - \min \Pr(A \cap B) = \frac{3}{4} + \frac{1}{3} - \frac{1}{12} = 1,$$

$$\min \Pr(A \cup B) = \Pr(A) + \Pr(B) - \max \Pr(A \cap B) = \frac{3}{4} + \frac{1}{3} - \frac{1}{3} = \frac{3}{4}.$$

Examples:

- Max 1: use the same $A = [0, \frac{3}{4}]$, $B = [\frac{2}{3}, 1]$ so the union fills $[0, 1]$.
- Min $\frac{3}{4}$: take $B \subseteq A$, so $A \cup B = A$.

Problem 8

Let C be the cheat die. Prior: $\Pr(C) = \frac{1}{9}$, $\Pr(C^c) = \frac{8}{9}$. Likelihood of two sixes: $\Pr(\text{two 6} \mid C) = 1$, $\Pr(\text{two 6} \mid C^c) = (\frac{1}{6})^2 = \frac{1}{36}$.

By Bayes,

$$\Pr(C \mid \text{two 6}) = \frac{\frac{1}{9} \cdot 1}{\frac{1}{9} \cdot 1 + \frac{8}{9} \cdot \frac{1}{36}} = \frac{\frac{36}{324}}{\frac{36}{324} + \frac{8}{324}} = \frac{36}{44} = \frac{9}{11}.$$

Problem 9

Each family has children until the first boy appears. Let $p = \frac{1}{2}$ be the probability a birth is a boy.

Number of boys per family: always 1 in this stopping scheme, so $\mathbb{E}[\text{boys}] = 1$.

Number of girls per family equals the number of failures before the first success in a geometric(p) model, so

$$\mathbb{E}[\text{girls}] = \frac{1-p}{p} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1.$$

Across the whole country, the expected counts scale with the number of families, giving the expected ratio

$$\text{boys} : \text{girls} = 1 : 1.$$

References